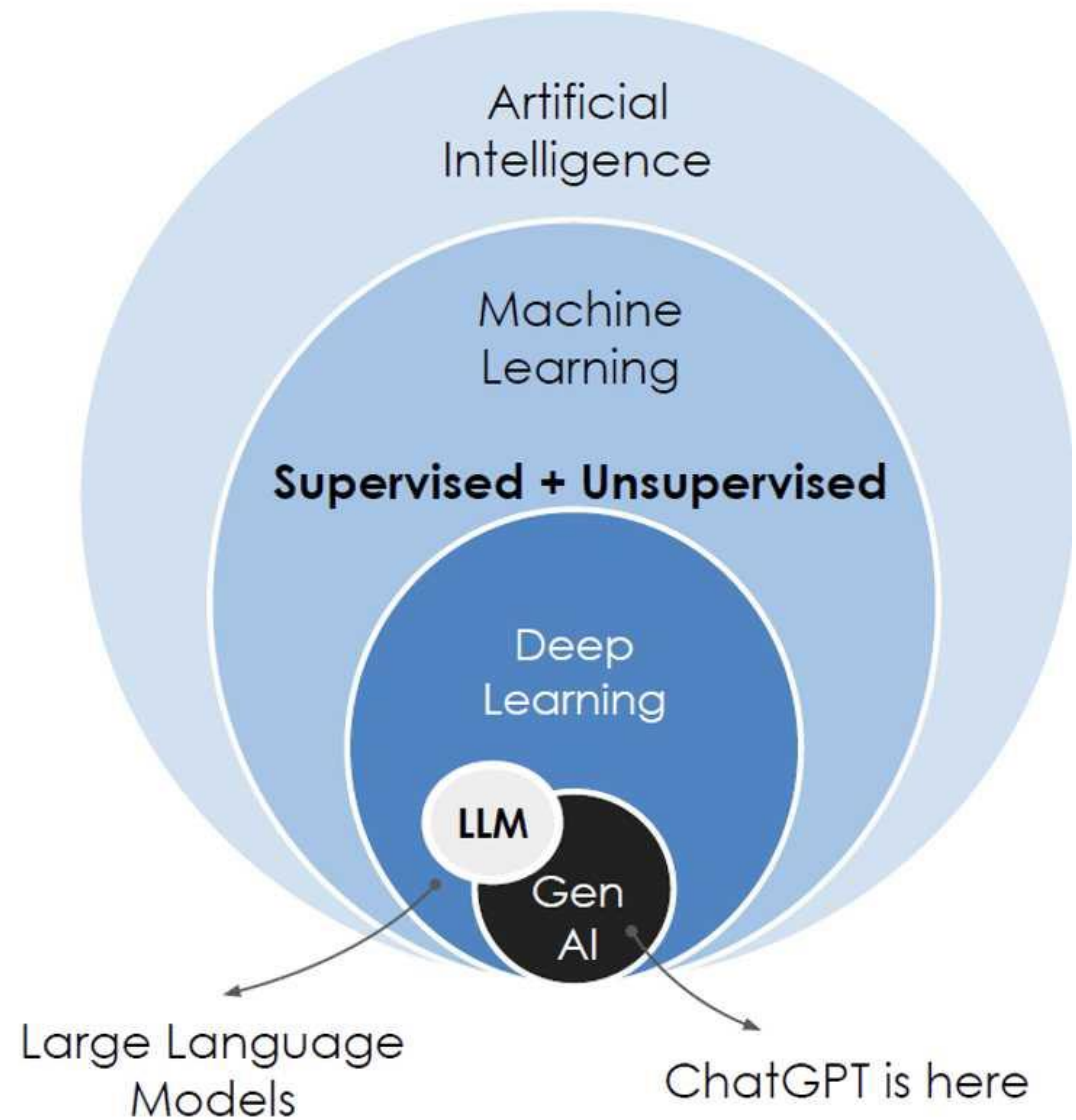


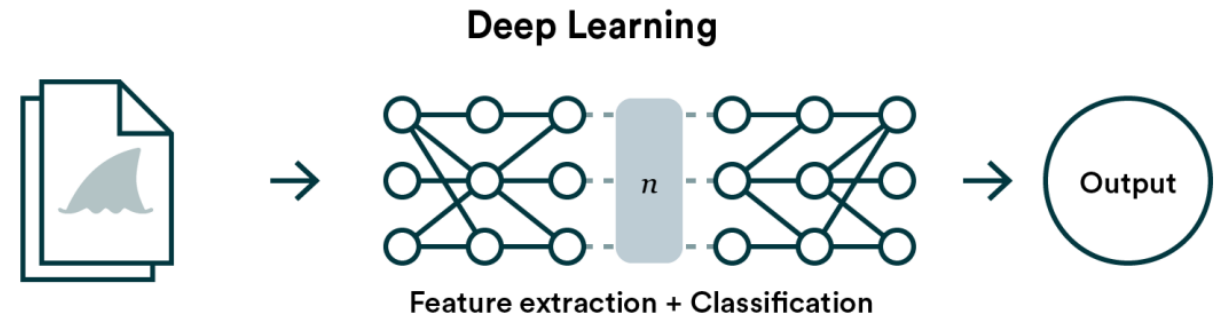
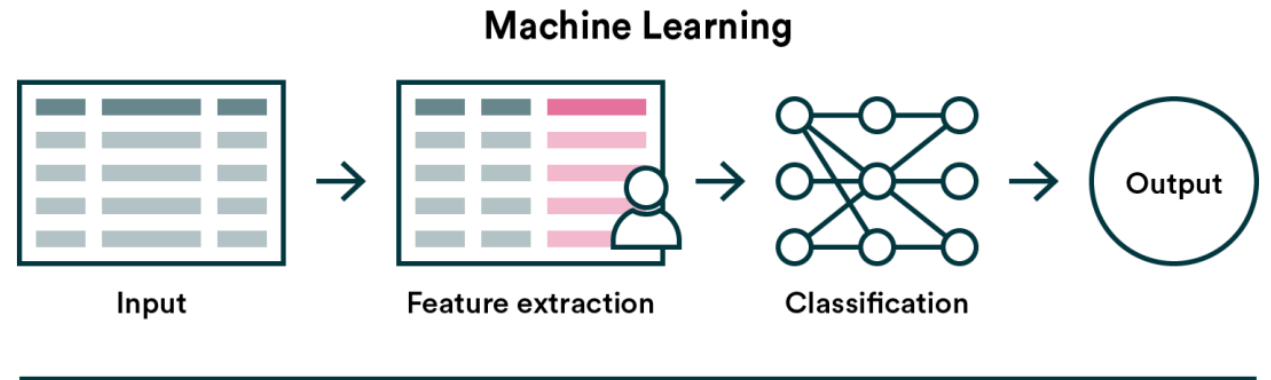
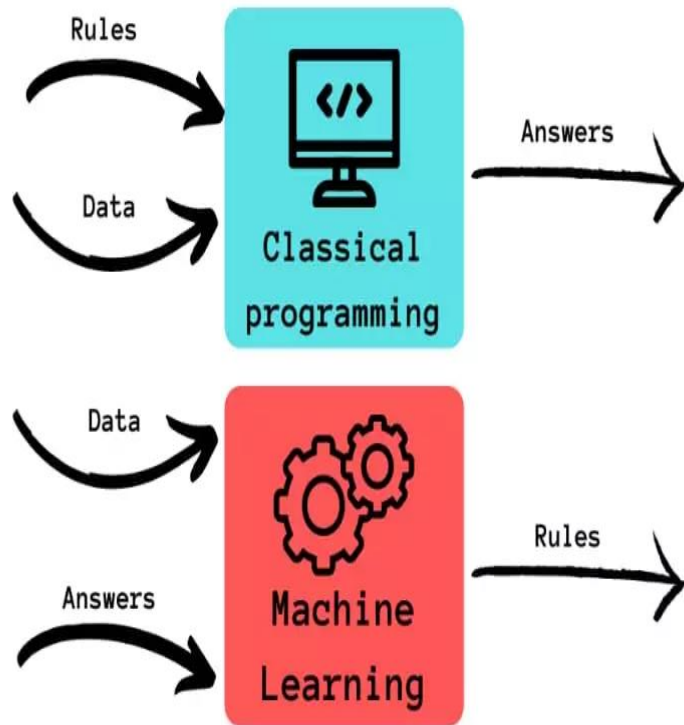
AI Foundations



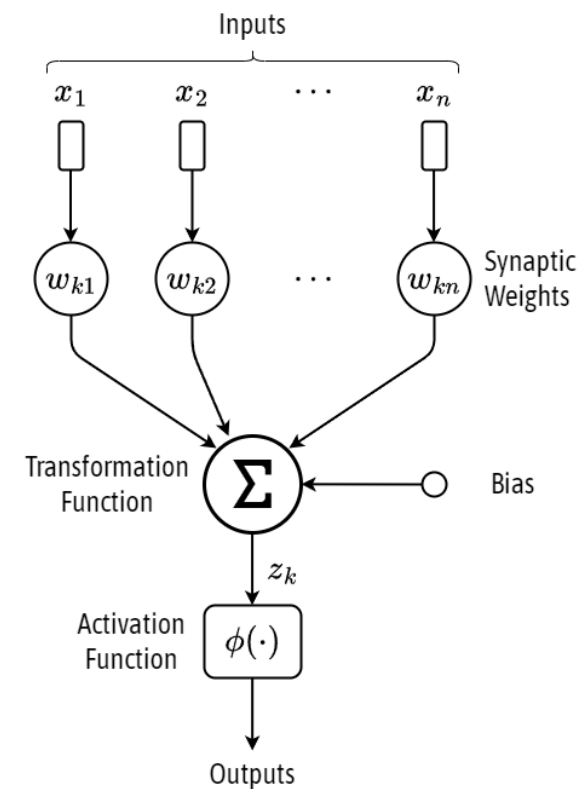
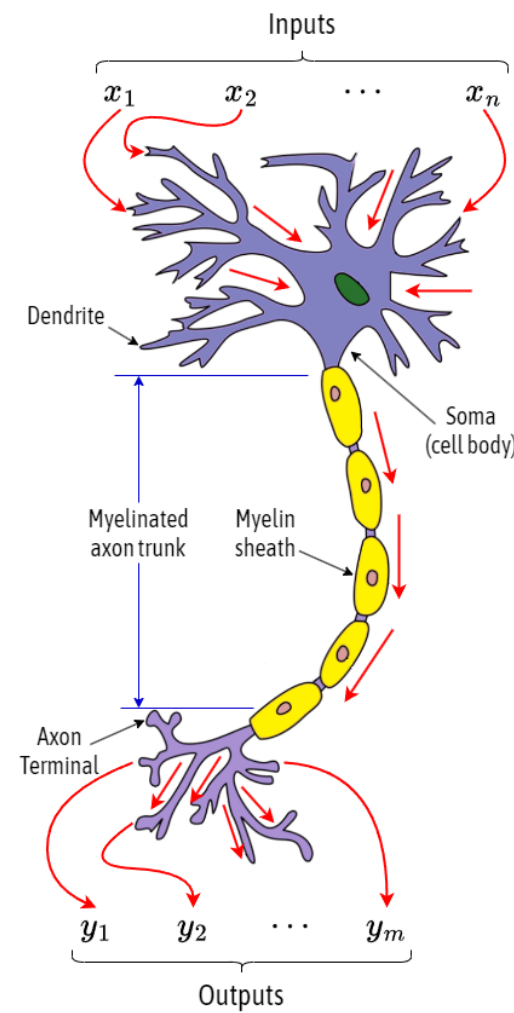
AI vs ML vs DL vs GenAI vs Agentic AI



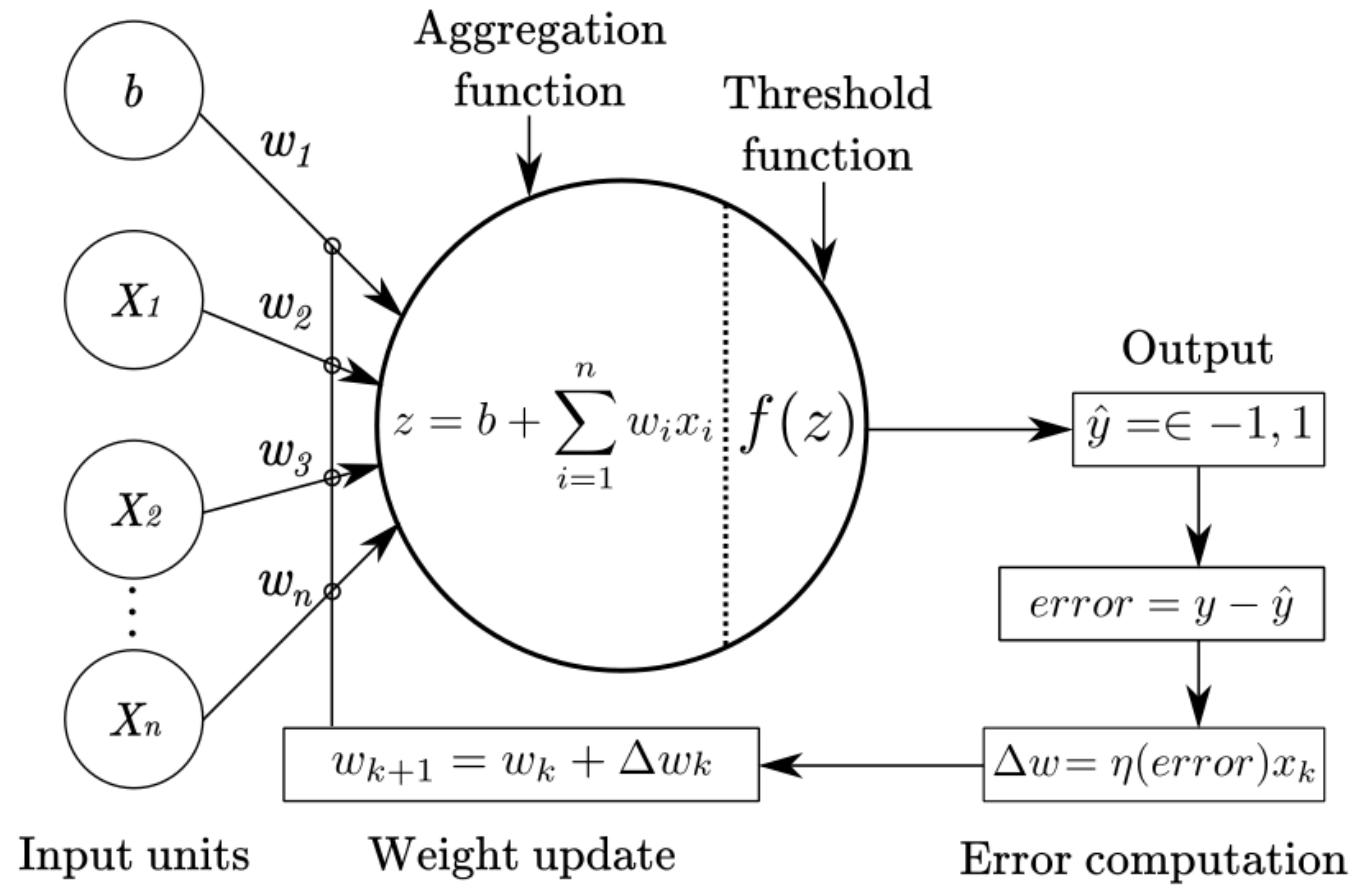
Evolution of AI



Neural Networks



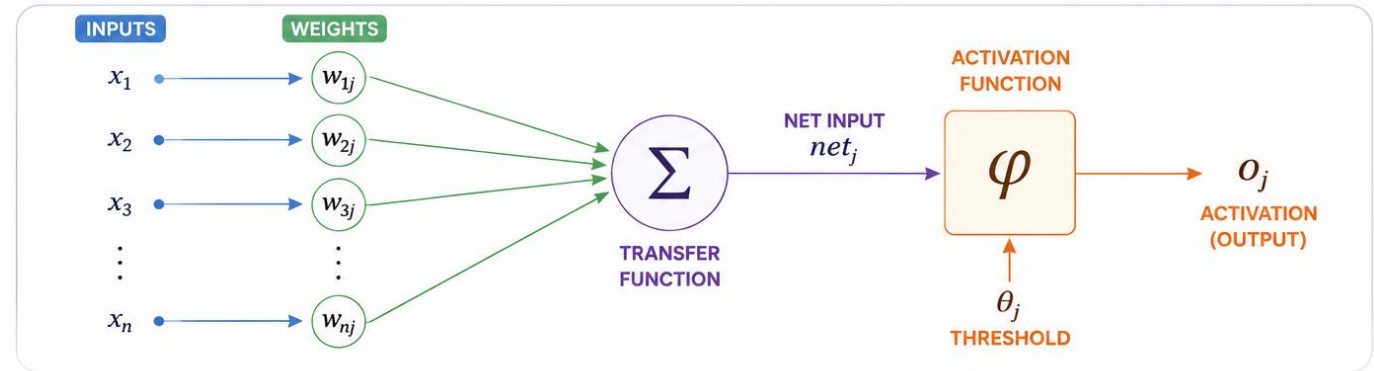
Closer Look



ANN for Tabular data

Artificial Neuron & Activation Functions

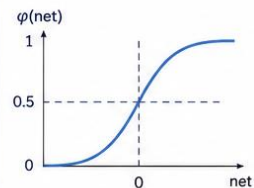
From Inputs to Output – The Heart of Neural Networks



1 TWO-CLASS: SIGMOID FUNCTION

Used for binary classification problems.

If threshold $> 0.5 \rightarrow$ One class
If threshold $< 0.5 \rightarrow$ Another class



$$\sigma(net) = \frac{1}{1 + e^{-net}}$$

Output range: (0, 1)
Common threshold: 0.5

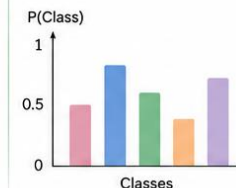


Example:
Spam Detection (Spam / Not Spam),
Disease (Yes / No)

2 MULTI-CLASS: SOFTMAX FUNCTION

Used for multi-class classification problems.

Gives probability for each class.



$$\text{softmax}(z_j) = \frac{e^{z_j}}{\sum_k e^{z_k}}$$

Output range: (0, 1)
Sum of all class probabilities = 1

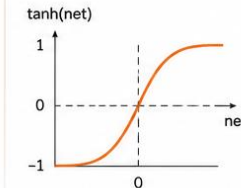


Example:
Digit Recognition (0-9),
Image Classification (Cat, Dog, Bird, ...)

3 TANH FUNCTION

Can be used as an activation function.

Outputs range between -1 and +1.



$$\tanh(net) = \frac{e^{net} - e^{-net}}{e^{net} + e^{-net}}$$

Output range: (-1, 1)
Zero-centered
(helps faster learning)



Example:
Hidden layers in Neural Networks,
Recurrent Neural Networks (RNNs)



Key Takeaway:

The transfer function (Σ) computes the weighted sum of inputs.

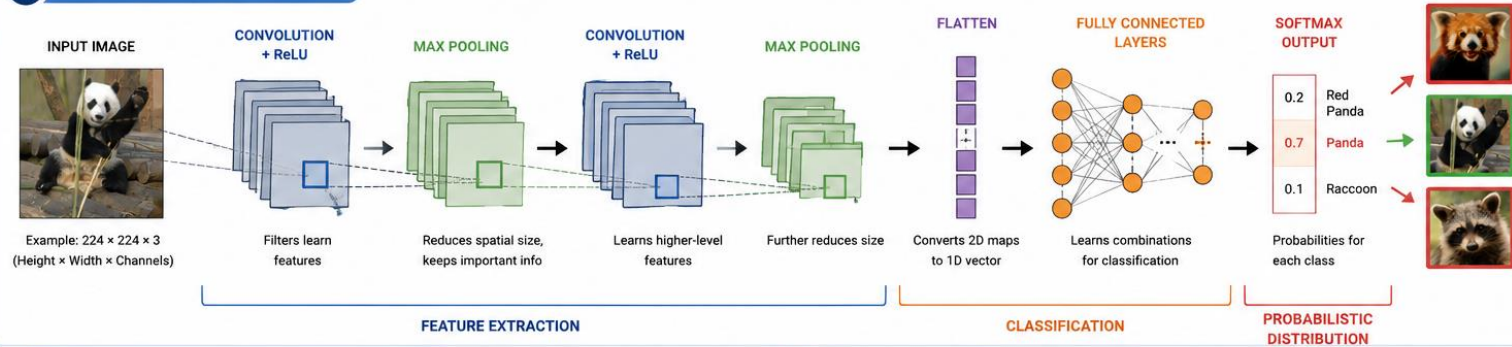
The activation function (φ) decides how that signal is transformed and passed forward.

CNN for Visual Data

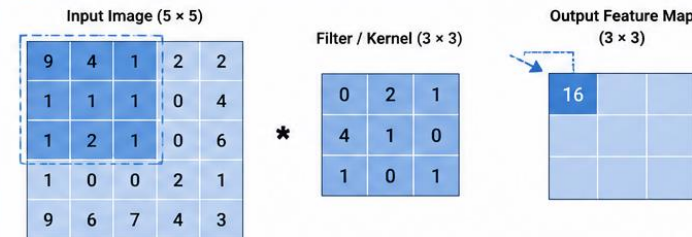
CONVOLUTIONAL NEURAL NETWORKS (CNNs)

From Pixels to Prediction

1 CNN ARCHITECTURE OVERVIEW



2 CONVOLUTION OPERATION (STEP BY STEP)

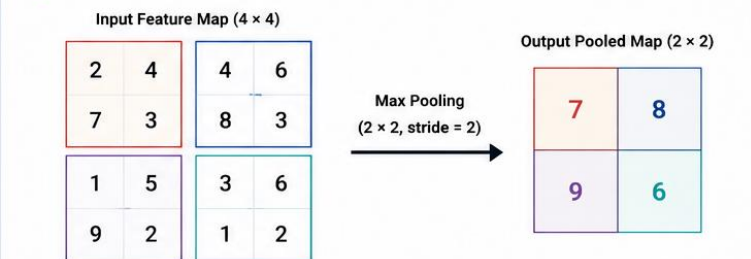


Calculation for Output[0][0] (Top-left corner):

$$(9 \times 0) + (4 \times 2) + (1 \times 1) + (1 \times 4) + (1 \times 1) + (1 \times 0) + (1 \times 1) + (2 \times 0) + (1 \times 1) = 16$$

- The filter slides over the image.
- At each position, element-wise multiplication and sum is computed.
- Produces a feature map.

3 MAX POOLING (DOWNSAMPLING)



- From each 2 $\times$ 2 region, the **maximum** value is selected.
- **Reduces spatial dimensions** while retaining the most important information.
- Helps **control overfitting** and reduces computation.



KEY TAKEAWAY

CNNs automatically learn hierarchical features from images using Convolutions and Pooling. The extracted features are then used by fully connected layers to predict the class with probabilities.

Convolution (Feature Learning)

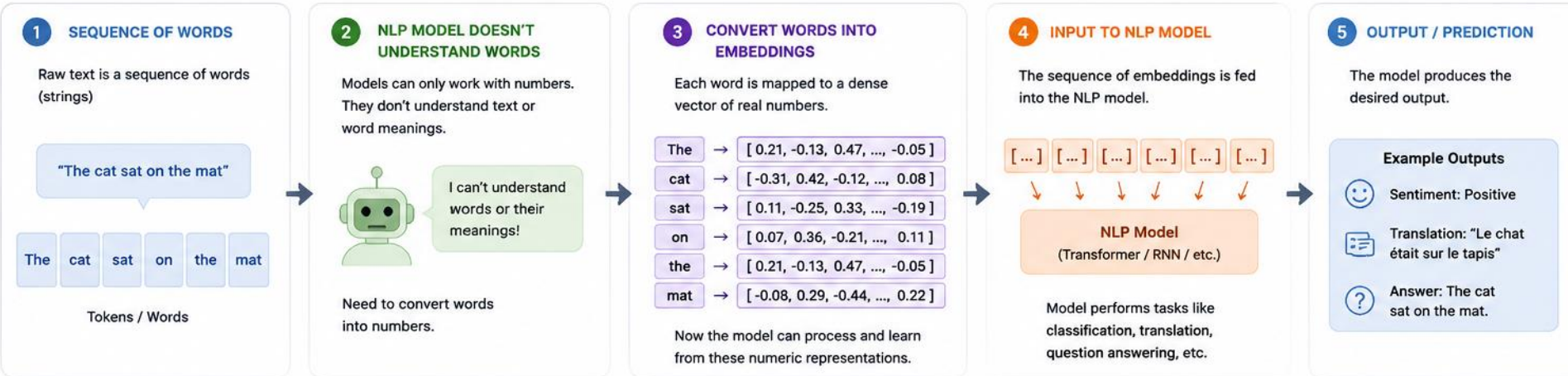
Pooling (Dimensionality Reduction)

Fully Connected (Classification)

Softmax (Probabilities)

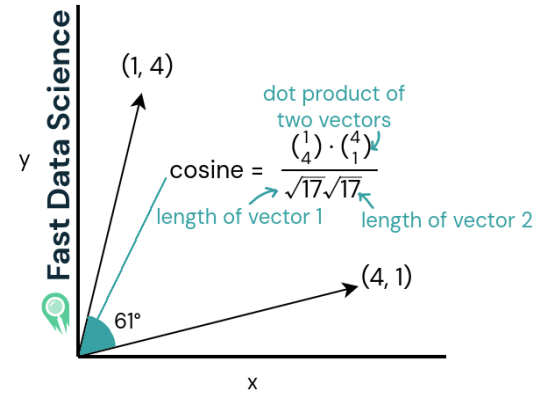
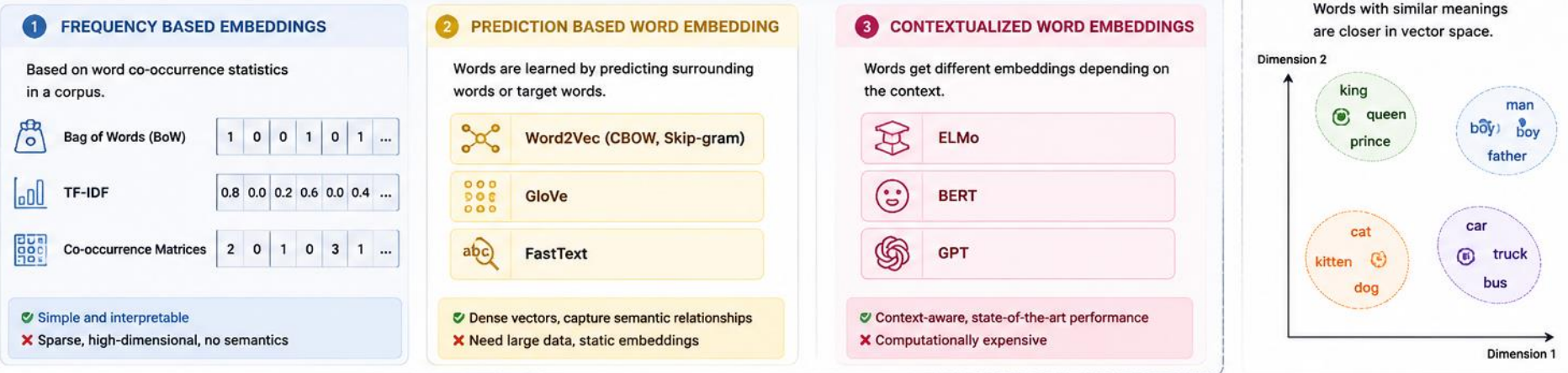
From Sequence of Words to NLP Model

NLP models don't understand words directly – they understand numbers (embeddings)



TYPES OF WORD EMBEDDINGS IN NLP

Embeddings represent words as vectors in a continuous space where semantically similar words are closer to each other.



cosine similarity is 0.47
this corresponds to angle of 61°

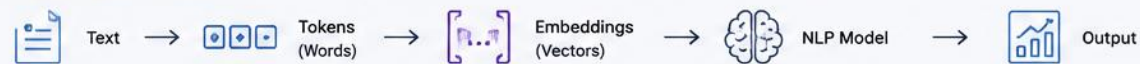
these two sentences would be far apart and not semantically similar



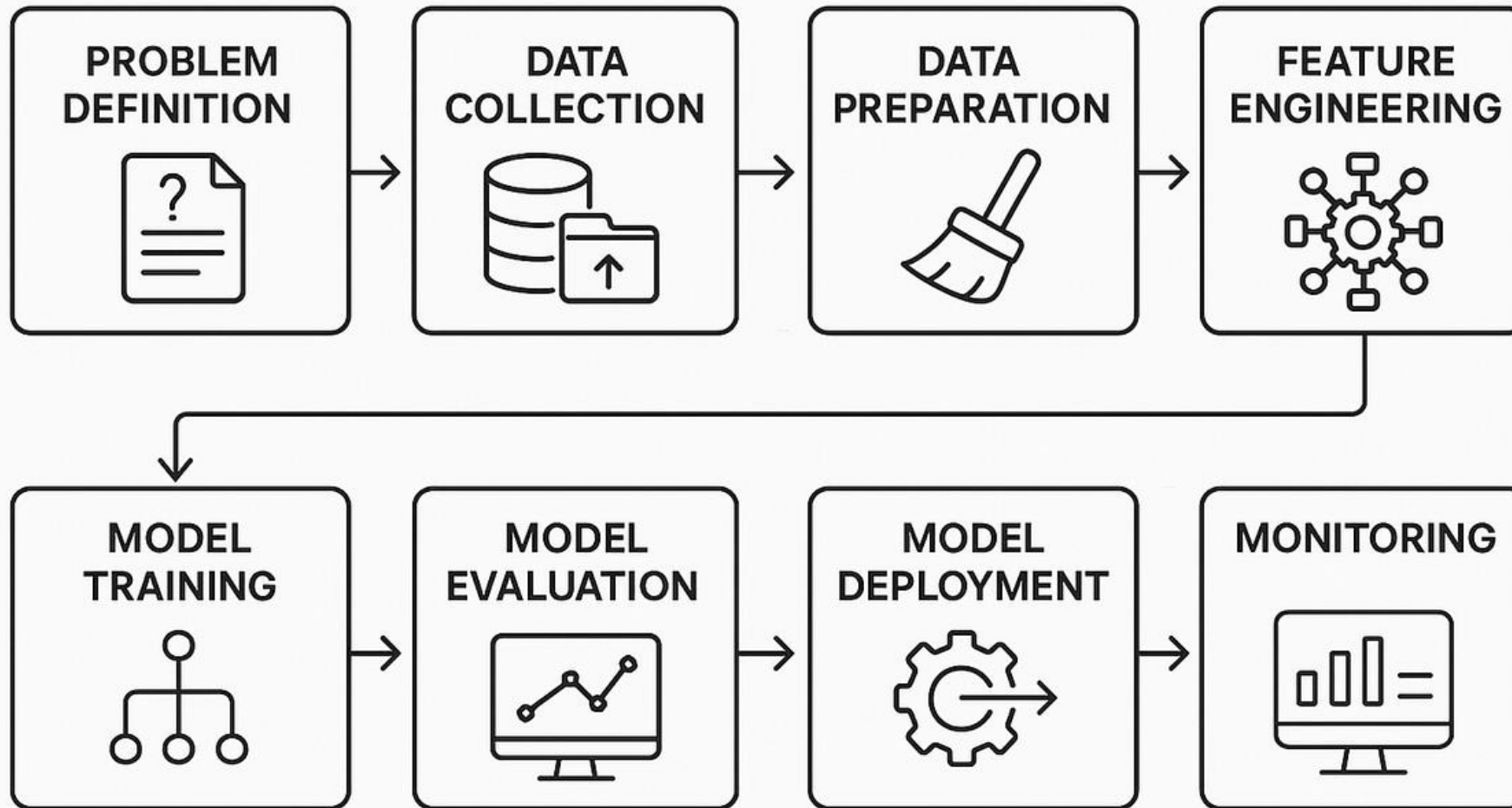
KEY TAKEAWAY

NLP models understand numbers, not words.
Word embeddings convert text into meaningful vectors, enabling models to learn patterns and make predictions.

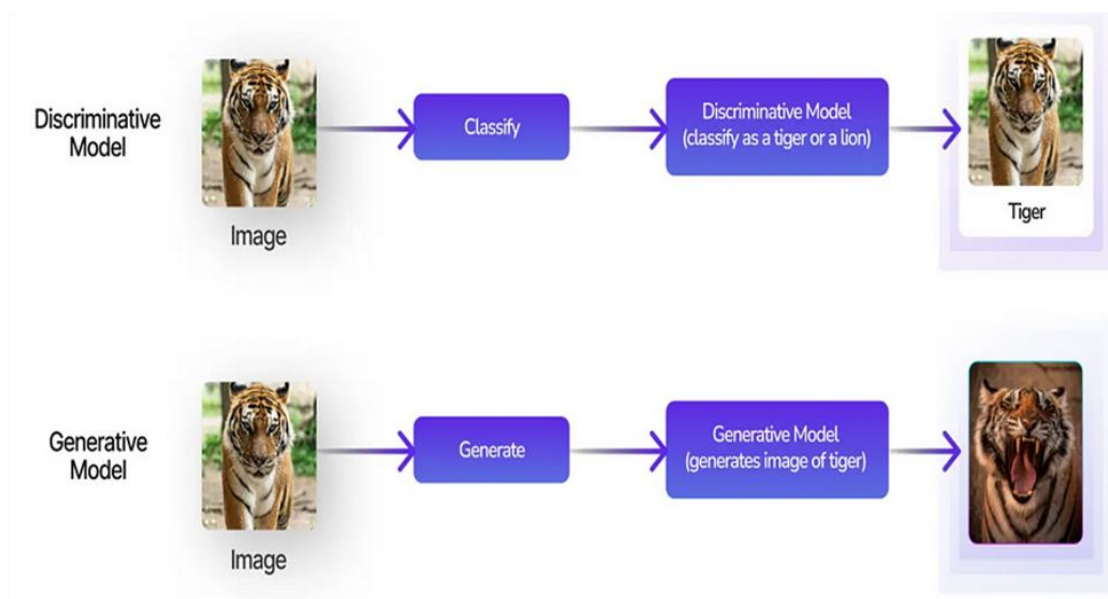
THE COMPLETE PIPELINE



AI LIFECYCLE

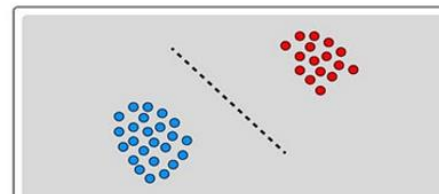


DISCRIMINATIVE VS GENERATIVE MODELS



Discriminative Models vs. Generative Models

Discriminative Models



Learns the decision boundary between classes

Maximizes the conditional probability: $P(Y|X)$

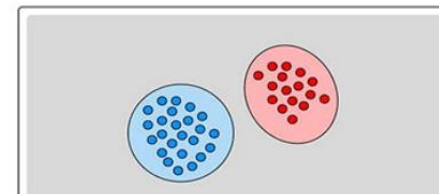
Directly estimates $P(Y|X)$

Cannot generate new data

Specifically meant for classification tasks

Discriminative models don't possess generative properties

Generative Models



Learns the input distribution

Maximizes the joint probability: $P(X, Y)$

Estimates $P(X|Y)$ to find $P(Y|X)$ using Bayes' rule

Can generate new data

Typically, they are NOT used to solve classification tasks

Generative models possess discriminative properties

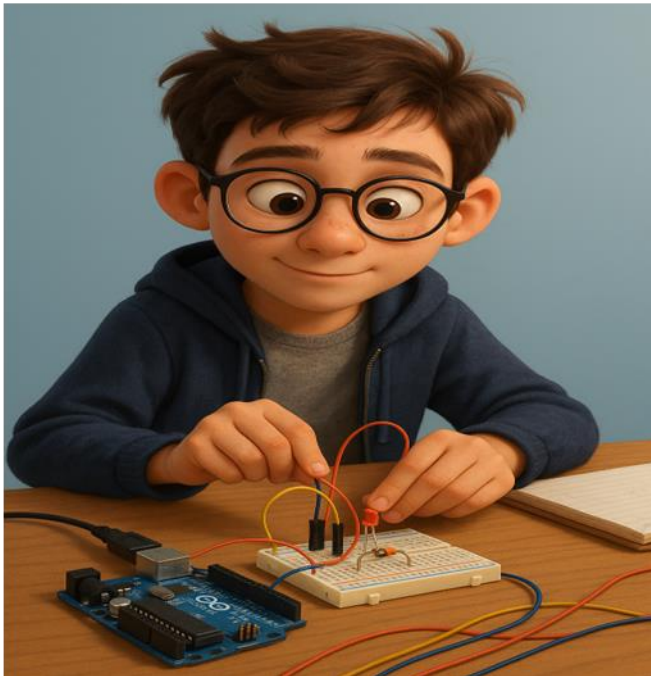
REAL-WORLD USE CASES - CHATBOTS

- **Type:** NLP (Natural Language Processing)
Technology Used: Large Language Models (LLMs) like GPT-4, BERT, LLaMA
- **How it Works:**
Generative AI-based chatbots process a user's prompt using a language model trained on billions of text documents. The model predicts the most likely next words based on context, generating coherent and human-like replies. They utilize transformers and attention mechanisms to understand context, tone, and intention.
- **Real Tools:**
 - **ChatGPT (OpenAI)** – Conversational assistant capable of answering questions, writing essays, solving problems.
 - **Google Gemini** – Multimodal assistant from Google DeepMind.
 - **Microsoft Copilot (Office 365)** – Embedded in tools like Word, Excel, and Outlook.



ChatGPT

IMAGE GENERATION (E.G., DALL·E, MIDJOURNEY)



- **Type:** Text-to-Image Generation
Technology Used: Diffusion Models, Transformers, GANs
 - **How it Works:**
The user provides a text **prompt**, such as “generate image where student working on Arduino”, and the model translates it into an image by learning the relationship between language and pixel patterns. Diffusion models start from noise and gradually denoise based on the prompt to generate the final image.
 - **Real Tools:**
 - **DALL·E 2 (OpenAI)** – Diffusion model-based image generator.
 - **Midjourney** – Highly aesthetic image generator via Discord bot.
 - **Stable Diffusion (Stability AI)** – Open-source image generation model.
-

TEXT-TO-VIDEO GENERATION - SORA



- **Text-to-Video Generation (e.g., Sora, Runway Gen-2, Pika Labs)**
- **Type:** Multimodal Generation (Text → Video)
Technology Used: Diffusion Models + Transformers + Temporal Modeling

Text-to-video models aim to generate **moving visual content (videos)** from natural language descriptions. The process involves:

1. **Text Understanding:** Using a transformer-based encoder (like in LLMs) to process the text prompt and extract semantic meaning.
2. **Frame Generation (Spatial Reasoning):** A diffusion or transformer model generates individual keyframes (images) from the prompt. This is similar to text-to-image models like DALL·E but with extra considerations for continuity.
3. **Temporal Coherence (Motion Modelling):** Ensures smooth transitions between frames using motion prediction modules. May use 3D convolution, optical flow estimation, or attention over time.
4. **Rendering and Upscaling:** The video is denoised, refined, and optionally upscaled to higher resolutions using tools like VideoGPT or Super Resolution models.

TEXT-TO-MUSIC GENERATION (E.G., MUSICGEN, SUNO AI, RIFFUSION)



- **Type:** Audio Generation (Text → Music)
Technology Used: Transformers, Diffusion Models, Spectrogram Inversion, Language-Audio Multimodal Models
- **How it Works:**
- Text-to-music generation converts a natural language prompt into an audio waveform that resembles a musical composition.
- Working Pipeline:
- **Text Encoding**
 - The input prompt (e.g., “Calm ambient background with piano and nature sounds”) is processed using a transformer-based text encoder to extract semantic features.
- **Audio Representation (Spectrograms)**
 - The model predicts a **spectrogram** (a visual representation of audio frequencies over time) or directly generates raw audio embeddings.
- **Audio Synthesis**
 - The spectrogram is inverted to waveform using tools like **Griffin-Lim algorithm** or neural vocoders such as **HiFi-GAN**.
- **Temporal Modeling**
 - Ensures musical patterns like rhythm, melody, and harmony remain coherent across time using autoregressive decoding or diffusion-based progression.

Nuance DAX Copilot



1. AI-powered clinical document solution from **Nuance Communications** which is a part of **Microsoft Cloud for Healthcare**, That uses **GenAI** to automatically **create clinical notes from patients' conversations**, reducing clinical burnout by automating administrative tasks, improving focus on patients.
2. **Ambient Capture** - Securely records multi-party conversations (clinician, patients and family) using mobile app at the point of care
3. **AI processing** - convert conversations into clinical summaries using Advance AI
4. **EHR Integration** - Integrate notes directly into the Electronic Health record
5. **Voice enable workflow** - Works with Dragon Medical One to automate tasks and create documents using voice commands

WHAT IS GENERATIVE AI?

Generative AI refers to a class of artificial intelligence models, especially the subset of Deep Learning, that **create new content**, such as **text, images, code, audio**, or even **3D models**, by learning patterns from existing data.

“While traditional AI classifies, detects, or predicts, Generative AI creates.”

Generative AI can produce:

- **Text**: Summaries, articles, emails, poetry (e.g., *ChatGPT, GPT-4*)
- **Images**: From simple text prompts (e.g., *DALL·E, Stable Diffusion*)
- **Audio**: Voice synthesis, music (e.g., *MusicGen, ElevenLabs*)
- **Code**: Autocompletion, generation (e.g., *GitHub Copilot, StarCoder*)
- **Videos** (new frontier): *Sora by OpenAI, RunwayML*



Generative AI Tools

Content Creation



Scribe



ChatGPT



Copy.ai



Jasper



Claude



Cohere



Gemini



Bard



Research Rabbit

Design & Visual Arts



DALL-E 2



Midjourney



Adobe Firefly



Canva AI

Coding & Development



GitHub Copilot



Turing's CodeGen Copilot



AlphaCode



Pico



Microsoft Copilot



Amazon Bedrock



Microsoft Power Apps

Audio & Video Generation



Synthesia



Auto-GPT



Audiovisual AI

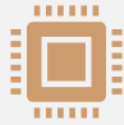


Elicit

Types of Generative AI Models



1. Generative Adversarial Network (GAN) - Generator vs Discriminator game



2. VAE (Variational Autoencoder) - Encoder-decoder architecture and Latent space sampling



3. Transformers - Attention mechanism, LLMs



4. Diffusion Models - Gradual denoising process



Thank You